

BUTACU IONEL-CATALIN

Machine Learning Engineer | Desktop & Enterprise Software Engineer

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SUMMARY

Software Engineer with 4 years of building desktop and enterprise software in C++ and Qt, plus machine learning engineering through graduate study, certifications, and ML projects. Built computer vision, RAG, and predictive systems covering model training, evaluation, inference, vector search, and edge deployment.

PROFESSIONAL EXPERIENCE

Gemini CAD Systems

Iasi, RO

Software Engineer

June 2023 – Present

- Built production C++/Qt features across UI, networking, and event-driven desktop modules.
- Built multi-instance crash recovery via shared-memory IPC and a dedicated crash-handler process.
- Integrated REST APIs and bidirectional cloud sync for core product features using Qt Network.
- Refactored legacy modules to reduce technical debt and improve maintainability.
- Resolved crashes, memory leaks, and regressions through root cause analysis.
- Contributed across Agile Scrum: planning, estimation, reviews, demos, and retrospectives.

Software Engineer Intern

August 2022 – May 2023

- Built desktop UI components from specs and integrated them into production views.
- Delivered pixel-accurate UI updates and extended real-time canvas functionalities.
- Worked across Git workflows, code reviews, and GitLab CI/CD.

Special Telecommunications Service (STS)

Botosani, RO

IT Technician – Electoral Bureau

August 2020 – September 2020

- Kept IT infrastructure operational across local electoral bureaus during the election process.
- Verified and transmitted electoral data through SIMPV and SICPV systems.
- Provided technical support and guidance to electoral bureau staff.

EDUCATION

Technical University of Iasi

Iasi, RO

Master of Science in Artificial Intelligence

October 2024 - June 2026

Advanced Coursework: Machine Learning, Deep Learning, Natural Language Processing (NLP), Reinforcement Learning, Multi-Agent Systems, Cloud Computing, Big Data Techniques, Edge Computing Applications, Intelligent Cyber-Security

Bachelor of Science in Systems Engineering

October 2020 - June 2024

Core Curriculum: Advanced Computer Programming, Data Structures & Algorithms, Real-Time Programming, Database Management Systems, Data Processing, Microprocessor-Based Systems, Embedded Systems Engineering, Control Engineering, Computer Networks, Numerical Methods, Neural Networks, Optimization Techniques, Fuzzy Logic, Knowledge-Based Systems

Independent Coursework: DataTalksClub Zoomcamps in Machine Learning Engineering and Data Engineering — Docker, MLflow, FastAPI, Terraform, Kubernetes, dbt, Spark, Kafka, and Airflow.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Data Science Professional (1Z0-1110-25) 
- Oracle Cloud Infrastructure 2025 Generative AI Professional (1Z0-1127-25) 
- Oracle Cloud Infrastructure 2025 AI Vector Search Professional (1Z0-184-25) 
- Oracle Analytics Cloud 2025 Certified Professional (1Z0-1041-25) 

COMPETENCIES

Software Engineering

- C++/Qt Desktop & Enterprise Development
- Software Architecture & Design Patterns
- Production Debugging & Root Cause Analysis
- Code Review, CI/CD & Agile Delivery
- Multithreading, Performance & System Design

Machine Learning & AI

- ML Model Training & Evaluation
- Agentic AI & Multi-Agent Systems
- Deep Learning & Computer Vision
- LLM Applications, RAG & Vector Search
- Model Deployment & Inference Serving

SKILLS¹

Generative AI:	HuggingFace 3.5 ★ · FAISS 3.5 ★ · BM25 3.0 ★ · CrossEncoder 3.0 ★
Machine Learning:	PyTorch 4.0 ★ · Scikit-learn 3.5 ★ · NumPy 4.5 ★ · Pandas 4.0 ★ · OpenCV 3.0 ★
Python Backend:	FastAPI 3.5 ★ · Pydantic 3.5 ★ · REST 3.5 ★ · gRPC 3.0 ★ · <i>pytest</i> 2.0 ★
C++ Stack:	Qt 4.5 ★ · MSBuild 4.0 ★ · STL 3.5 ★ · CMake 3.5 ★ · SonarQube 4.0 ★
Languages:	Python 4.0 ★ · C++ 4.5 ★ · SQL 3.5 ★ · Bash 3.5 ★ · TypeScript 3.0 ★
DevOps & Cloud:	Docker 4.0 ★ · Linux/WSL 4.0 ★ · Git 4.0 ★ · OCI 3.5 ★ · AWS 3.0 ★ · Terraform 2.5 ★
Data Engineering:	<i>Kafka</i> 2.0 ★ · <i>Airflow</i> 2.0 ★ · <i>Spark</i> 2.0 ★ · <i>dbt</i> 2.0 ★

PROJECTS

Text2Motion with Physics-Constrained Generation (<i>MSc Dissertation, in progress</i>)	2026
Generative AI pipeline converting natural language prompts into realistic 3D motion sequences. Modular Python architecture: MoMask generative transformers for motion synthesis, SBERT for semantic parsing of input prompts, and a PyBullet physics engine enforcing biomechanical constraints and environmental interactions.	
IDS/IPS Smart Firewall	2026
Collaborative intelligent firewall system featuring a CNN threat classifier and an RL adversarial agent. Personal contribution focused on system architecture, Snort rule configuration, and developing a DBSCAN clustering module that analyzes runtime payloads to auto-generate rules, reducing ML inference load. Deployed on Ubuntu Server / Kali Linux.	
Hand Gesture Recognition	2025
YOLOv5n object-detection pipeline for hand gesture recognition (HaGRID dataset). Optimised for NPU/CPU deployment via TFLite INT8 quantisation, CLAHE contrast preprocessing, and temporal smoothing to reduce false positives. Developed and benchmarked on Android OS for NXP i.MX.	
Med Assist — Romanian Pharmacy AI Assistant	2025–2026
Full-stack RAG chatbot over the Romanian official ANMDM drug registry (7,555 entries). React + TypeScript frontend; FastAPI + Pydantic + Postgres backend. Hybrid retrieval combines FAISS dense embeddings, BM25 sparse search, and a cross-encoder reranker; Google Gemini generates the final response, with a safety layer escalating severe symptoms to professional care. Auth0 JWT, Docker, Terraform-managed cloud infrastructure.	
Metaheuristic Framework	2025
Python library implementing Canonical GA, Real-Valued GA, Differential Evolution, and Random Search. Each solver is benchmarked on standard mathematical test functions with automated CSV reporting and convergence visualisation.	
fMRI Brain-Age Prediction	2025
Competition ML pipeline predicting biological brain age from ~20k fMRI neuroimaging connectivity features. Gender-stratified ElasticNet model with spectral feature selection; ranked 8th nationally at the WiDS Datathon 2025 University Edition.	

VOLUNTEER EXPERIENCE

LOCAL BEST Group Iași	Iași, RO
IT Department Coordinator	Sep 2023 – Aug 2024
- Directed IT operations and maintained technical infrastructure across organisational events. - Mentored volunteers and oversaw delivery of technical projects. - Administered hardware, software licensing, and network resources.	
Event Coordinator JobShop Career Event	Nov 2022 – Apr 2023
- Coordinated logistics and operations for a large student–industry career event. - Managed scheduling, partner engagement, and on-site execution. - Improved attendee experience through strategic planning and workflow optimisation.	
TOPICS Responsible Prove It 24H Techathon	Dec 2021 – Mar 2022
- Defined technical requirements and coordinated hackathon logistics. - Supported participants with rules, resources, and real-time assistance. - Facilitated communication between organisers, industry mentors, and teams.	
IT Responsible BEST Training Weekend	Jan 2021 – May 2022
- Developed and maintained the event website using HTML, CSS, and JavaScript. - Collaborated with design and marketing responsible members on content, branding, and technical setup.	

¹ Scale (out of 5): 5 owned in production 4 strong hands-on use 3 comfortable, still learning 2 coursework-level / guided exposure (*italicised*) 1 foundational awareness